

V2V-Based Memetic Optimization for Improving Traffic Efficiency on Multi-Lane Roads

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Abstract—Within the next few years, autonomous vehicles will start being commercialized. In the same time, inter-vehicular communication is emerging. Using these new technologies allows designing new software to make cooperative cars. These cooperatives cars can exchange messages to improve traffic efficiency by, for instance, notifying about the presences of other cars, or managing the right-of-way at intersections. In the context of multi-lane roads, we propose to use the communication between vehicles to design a cooperative intelligence, based on evolutionary optimization where the behavior of each vehicle is regularly updated according to the behavior of surrounding vehicles and a fitness function. This paper presents an automated/cooperative lane-change framework where the parameters of the system are dynamically adjusted using an online evolutionary algorithm. The goal is to make the cars adjust their behavior according to the local traffic conditions. Simulations are carried out showing a performance improvement in terms of traffic fluidity.

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I. Introduction

More and more cars are being equipped with advanced driver assistance systems (ADAS), like adaptive cruise control (ACC) which help the drivers to respect a given headway with preceding cars, or automatic lane-change taking care of overtaking and merging operations. Simultaneously cars are beginning to be equipped with vehicular communication units or on-board units (OBU) allowing them to communicate with the surrounding cars (V2V) or with the infrastructure (V2I).

More than a decade ago, the use of communication between vehicles inspired studies on cooperative traffic where autonomous or semi-autonomous cars can cooperate. One of the forms that this cooperation can take is the Cooperative Adaptive Cruise Control (CACC) where vehicles adapt their speeds using received wireless messages instead of the data provided by the sensors, thus improving the performance of classical ACC systems [1], and improving the traffic flow [2]. Another form of the cooperative traffic is the cooperative intersection management (CIM) where cars are using wireless communication to get the right-of-way at intersections, and thus reduce the time spent in the intersection.

CIM was introduced in the late 90 s [3] and it is of growing interest due to the current advancements in driverless vehicles. Preliminary works introduced an on-board display to replace traffic lights at intersection, with a server in charge of managing the junction and giving the right-of-way to the different lanes. Furthermore, it allowed individualizing the right-of-way on a per car basis instead of on a per lane basis, authorizing more complex sequences of passage, and thus enabling the optimization of the sequence of vehicles [4]. Finally, using autonomous vehicles permits to optimize both the sequence of passage and the speed of vehicles, by synchronizing the vehicles speeds to reduce the time spent in the box junction [5, 6]. Successful experiments have been conducted at the Intelligent Transportation Systems World Congress in Bordeaux in 2015, using standard robotized vehicles with communication units as displayed in the figure 1).



FIG 1 Two autonomous cars synchronizing their speed at an intersection using Vehicle-to-Infrastructure communication at the ITS-WC 2015 in Bordeaux [7].

Since the release of these works, the theory and the technology have improved, and current works are targeting to improve the quality of the communication [8], which extend the possibilities of cooperation. Thus, nowadays the cooperative traffic management is being extended to all the traffic operations which could imply or be improved by a cooperation between several agents. These works are often gathered under the name of “Cooperative Traffic Management” (CTM) or “Cooperative Traffic Control” (CTC) and various strategies are designed to improve the traffic efficiency by using the communication. In this domain, many interesting works are being developed like [9] and [10].

Recently several works are extending the studies on the CIM to the case of lane-change [11], given that the merging operation can be seen as an intersection where the conflict area (the area that will be traversed by two vehicles) is moving. Thus, the cooperative lane-change (as defined in [12]) can be automatized. It introduces the possibility to optimize the traffic with communication. Autonomous cars can optimize lane-change operations with speed synchronization, and can adjust their behavior (target speed, accepted deceleration, etc.) to ease the travel of the other vehicles. Although [13] and [11] works are restricted on a single intersection, it already shows benefits of using cooperation for merging and lane-change operations.

Furthermore, in the context of multi-lane roads, the behavior of a car can be characterized by a set of internal parameters, like the target speed, the headway time, etc. which have an influence on the overall traffic performance.

If the assumption that the position of all cars cannot be perfectly known in real-time is made, defining the ideal value for these parameters to optimize the traffic performance (both debit and fluidity) is a complex task. In terms of game theory, it is a continuous game with imperfect and incomplete information. Thus, it is difficult to estimate how performing a lane-change, or setting a parameter for an individual car will affect the traffic according to the context. Moreover, the traffic conditions are always changing (density, speed limits, number of lanes, etc.) and statically setting cars parameters does not provide a solution able to adapt to these changes. So, to be applicable, the optimization process must be made online.

Thus, using the idea that if the internal parameters of a car positively affect its performance in the traffic, it is interesting to imagine how communication could be used to propagate the “good” behaviors to the surrounding cars. Fortunately the literature has a concept for this transmission of a behavior from person to person within a culture: the meme [14]. This concept states that behaviors evolve in a manner analogous to that of biological evolution, through processes of variation, mutation, competition and inheritance. Reused by the computer science, this leads to the construction of a class of evolutionary algorithms called “memetic algorithms”.

The purpose of this article is to present a framework that can be used to optimize highway driving (which includes multiple and simultaneous lane-change operations, frequently causing slowdowns). Indeed, even though a lot of research has been produced on the automation of lane-change [15, 16], these works are usually focusing on the control of the vehicle rather than the possibilities of optimization introduced by the cooperation.

This article details the context of cooperative lane-change, and then proposes a framework wherein autonomous cars are able to communicate and cooperate to dynamically adjust their parameters to improve overall traffic performance. The objective is to optimize highway driving. Thus, a framework is based on the principles of memetic optimization is proposed, where an evolutionary algorithm (at network level) is used in combination with a local search algorithm (at vehicle level). The purpose of using a memetic algorithm here is not to perform a classical offline optimization, but is to serve as a basis to make a swarm intelligence emerge from the behavior of the vehicles. Other stochastic algorithms could be used but as we will see later, the use of a local-search part relying on heuristics greatly improve the overall performance, and led us to the use of a memetic algorithm.

II. Cooperative Lane Change

As explained sooner, the goal of the presented framework is to allow the optimization of multiples and simultaneous operations of lane-change on the highway. As a basis, the principles of Cooperative Lane Change (CLC) are used. CLC is the extension of the CIM principles to the case of lane-change, and thus, allows introducing cooperation on a single lane-change operation. This section details the CLC protocol, which composes the multiple cooperative lane change (MCLC) protocol.

A. Definitions

In the following, the definitions displayed in the figure 2 and detailed in the table 1 are used to refer to the cars involved in the lane change operation. These definitions are similar to the ones in [17].

B. CLC Protocol

In this article, a CLC protocol based on the Transparent Intersection Manager (TIM) is used [6]. TIM builds an ordered sequence of passage of vehicles for a conflict area and allows cars to synchronize their speed with the preceding vehicle in the sequence. Adapted to lane-change, it results in the algorithm 1.

Although the global protocol is simple, it relies on several models and algorithms to perform the required operations.

The function *isLaneChangeDesired()* relies on a lane-change model which defines if a lane-change must be performed. For instance, the model defined in [17].

The function *isLaneChangePossible()* uses radar and a longitudinal model to ensure that a lane-change can be safely performed, and that it will not make the new follower decelerate above a given limit (0 m.s^{-2} to avoid any deceleration). The radar measures the relative position of the other car, and the longitudinal model estimates its expected acceleration or deceleration. For instance, in the

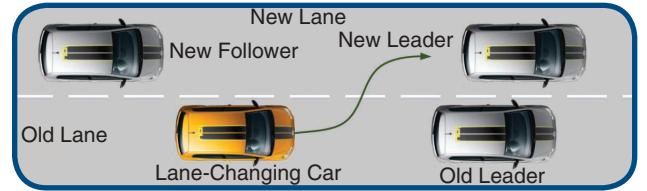


FIG 2 Lane change definitions.

Table 1. Terms and definitions of lane change.

Terms	Definitions
Lane changing car	The car which is making a lane- change
Old lane	The current lane
New lane	The lane targeted by the lane changing car
Old leader	The current leader, i.e. the leader of the lane changing car before the lane change
New leader	The leader of the lane changing car in the new lane, i.e. the closest car in front of the lane changing car which is in the new lane
New follower	The follower of the lane changing car in the new lane, i.e. the closest car behind the lane changing car which is in the new lane
Intersection point	The point where the lane changing car will switch to the new lane

Algorithm 1 Cooperative Lane-Change protocol

```

if isLaneChangeDesired() then
  if isLaneChangePossible() then
    performLaneChange()
  else
    intersection ← defineIntersection()
    cooperativeVehicles ← []
    repeat
      broadcastIntersectionData(intersection)
      getCooperativeVehicles(cooperativeVehicles)
      synchronizeSpeed()
    until isCLCPossible(cooperativeVehicles) or
      !isLaneChangeDesired()
    if isCLCPossible(cooperativeVehicles) then
      performLaneChange()
    end if
    broadcastIntersectionEnd(intersection)
  end if
end if

```

MOBIL model [18], the Intelligent Driver Model (IDM) [19] is used to define if a lane-change can be performed.

performLaneChange() relies on a trajectory planning model and a lateral control solution, to control the steering wheel angle, and thus performing the lane-change.

If the lane-change procedure is not possible according to radar (i.e. *isLaneChangePossible()* has returned *false*), then the cooperative lane-change procedure begins. A virtual intersection object is defined, which contains data about the involved lanes, a limit point where the lane-change must have been performed, and the internal data of the lane-changing vehicle (at least position and speed). This information is regularly updated and broadcasted to surrounding cars, which may answer if they accept to let the priority to the lane-changing car at the intersection (thus adding them to the list *cooperativeVehicles*). Cooperative vehicles will adjust their speed to create a gap with the preceding vehicle which will allow the lane-changing car to merge in their lane. Moreover, cooperative vehicles emits their position, speed and the relative position of the preceding car, thus enabling speed synchronization of the lane-changing car: *synchronizeSpeed()* synchronizes the speed of the vehicle with the closest cooperative car.

Finally, if the cooperative lane-change is possible, i.e. the lane-change can be safely performed without making the new follower decelerate above its comfortable deceleration value (usually lower than the previous deceleration limit, for instance -2 m.s^{-2} , given that the new follower agreed to let the lane-changing car merging in its lane), then the lane change is performed.

The global state-chart of the system is detailed in the figure 3, including the lane-change part as detailed in the algorithm 1, but also the synchronization part defining how the other cars may cooperate with the lane-changing cars. We draw the reader attention to the fact that compared to *isLaneChangePossible()*, *isCLCPossible()* does not introduce any additional risk of collision since the overtaking decision still relies on radars. However, it improves the overtaking condition by slowing down the new follower.

Therefore, the architecture presented in the figure 4 is required to implement CLC. It can be noted that it is similar to the architecture of CIM, except that a lane-change planner and a lane-change model are required.

Simulations have already shown that the use of CLC on an automated highway can improve the traffic fluidity in specific cases [20].

III. Problem Statement

In the CLC protocol previously detailed, numerous parameters can be adjusted, about the vehicle (target speed, comfortable deceleration, etc.), or about the interaction between the vehicles (accepted deceleration, distance to intersection point, etc.).

The selection of values for this parameter will have a direct influence on the overall traffic performance. Moreover, different traffic conditions may require a different set of parameters to optimize the traffic. The present problem is to build a solution to define and dynamically adjust those parameters.

A. Assumptions

For this study, the problem is restricted to highway driving. It means that there are no blind intersections (vehicles are all able to perceive surrounding cars). The supposition is made that the vehicles are all autonomous for highway driving, i.e. able to adjust their speed according to the data perceived by the sensors, able to perform lane change operations and able to communicate using V2V.

A more realistic hybrid system, with both autonomous and standard vehicles, is possible. The only difference is that the optimization process would not be applied to the standard vehicles. Therefore, for the sake of simplicity all vehicles are assumed autonomous in this article.

B. Parameters

Every vehicle has a predefined target speed that it will try to respect. Thus, the objective of our framework is to minimize the average difference between the effective speed v of a vehicle (which depends on surrounding vehicles) and its target speed v_i .

For the longitudinal model of the vehicle, RT-ACC is chosen [21] as it is specifically designed for speed synchronization using V2X. For the lane-change model of the vehicle, a MOBIL-like model is used (detailed in 5.1).

The parameters of vehicle are defined in the table 2.

In the following, any occurrence of one of these symbols will refer to the associated definition.

C. Performance Evaluation

To evaluate the performance of the framework, three measures are made:

- 1) the average speed of vehicles on the highway $\bar{v}$
- 2) the average difference between the effective speed of a vehicle and its target speed $\bar{\Delta v}$
- 3) the index of fluidity (IF) defined by equation 1 [22], with:
 - a) q_i the number of vehicles having traveled on the section i
 - b) l_i the length of this section
 - c) v_i^f the “free” speed on this section (i.e. the average target speed in our case)
 - d) v_i the effective speed on this section

$$IF = 10 \times \frac{\sum_{i=1}^n \frac{q_i l_i}{v_i^f}}{\sum_{i=1}^n \frac{q_i l_i}{v_i}} \quad (1)$$

IF is used to measure the fluidity of the traffic, by dividing the time spent at design speed by the time spent

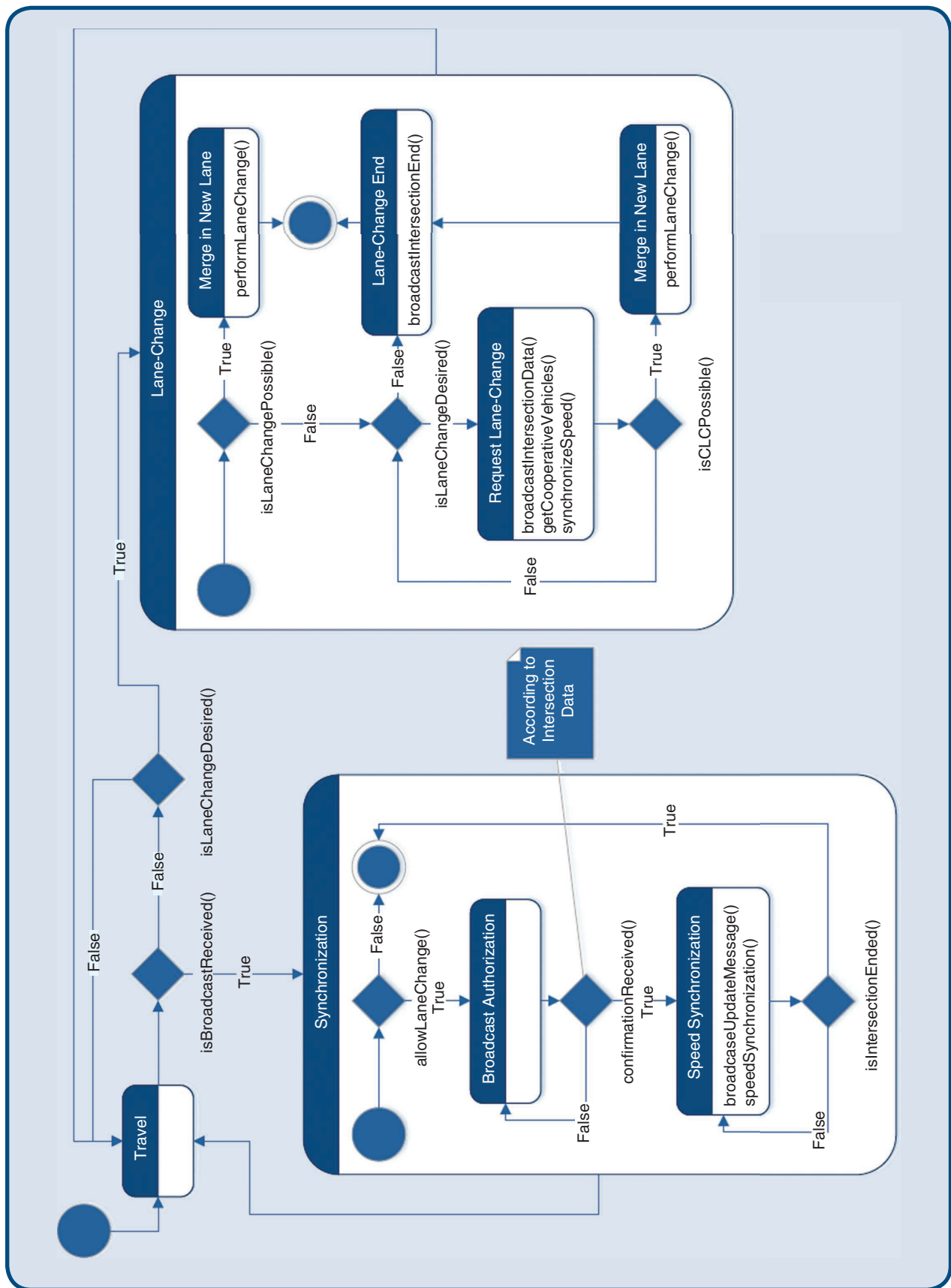


FIG 3 Global state-chart of the Cooperative Lane-Change system.

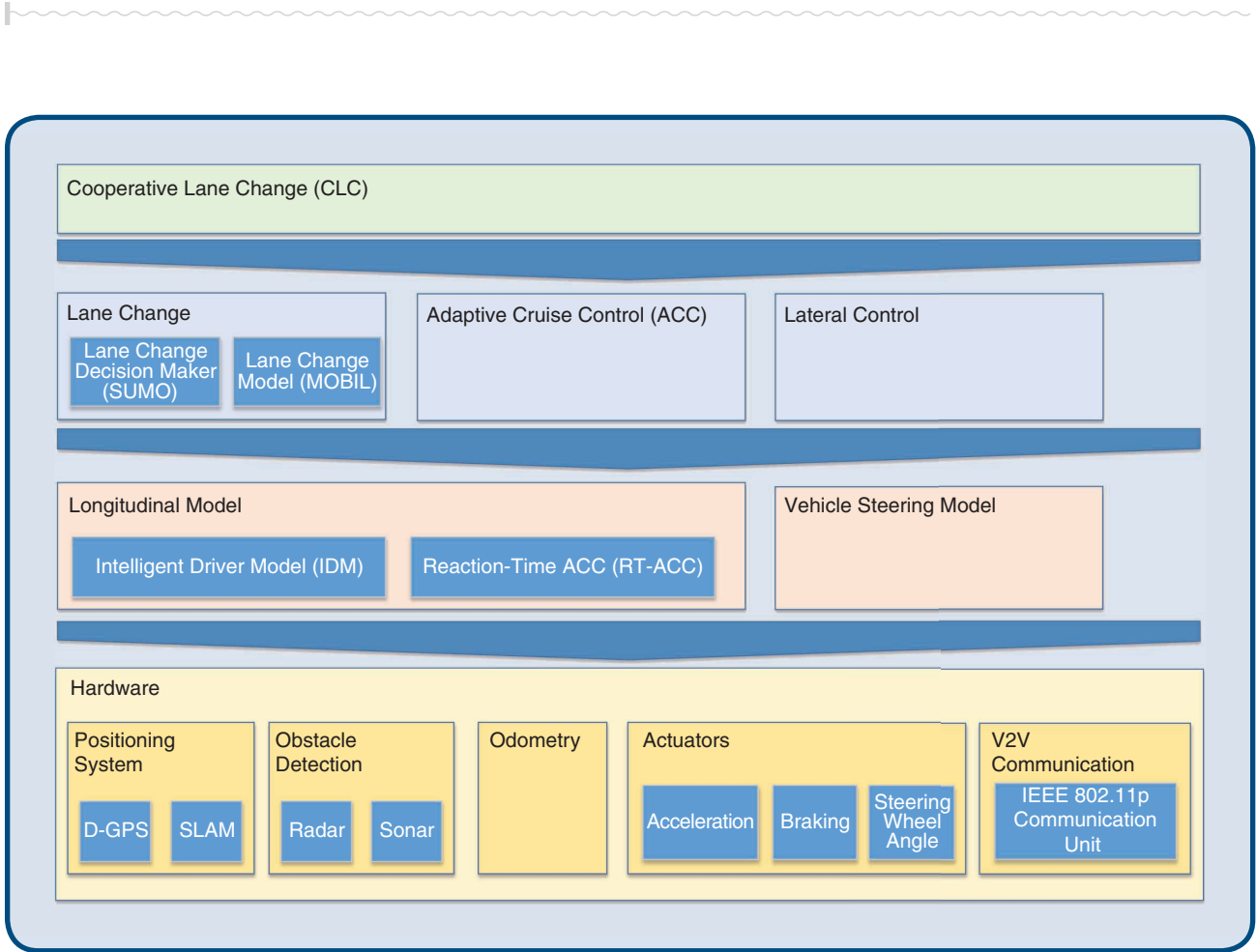


FIG 4 Architecture of the CLC system.

Table 2. Vehicle parameters in the context of cooperative lane change.

Parameter	Meaning
v_t	The target speed
b_t	The preferred deceleration, i.e. the deceleration value that a car will not exceed, except in case of immediate danger
τ	The tau, which is used here as an equivalent of the headway time but for RT-ACC
Δ_a	The minimum delta acceleration, i.e. the minimum benefit in acceleration required to perform lane-change
b_r	The maximum imposed deceleration using radar, i.e. the deceleration value that can be imposed to a car when performing a lane-change (for instance a value of $0\text{m}\cdot\text{s}^{-2}$ means that a car will not perform a lane-change if it would make a car in the other lane decelerate)
b_v	The maximum imposed deceleration using V2V (overrides b_r when cooperative lane change is enabled)

at actual speeds. According to the expression of the IF, a value of 0 means that the traffic is blocked (because IF is equal to 0 when $v_i \rightarrow 0$) and a value of 10 means that the traffic is fluid (because IF is equal to 10 when $v_i \rightarrow v_i^r$).

IV. Cooperative Intelligence

The problem is inherently distributed: the goal is to improve the traffic performance, and it can be made by modifying the behavior of each individual car.

Therefore, a multiagent system (MAS) is an appropriate tool for modeling the system, where each vehicle is an agent, sending messages to the surrounding vehicles through V2V communication. In this system, each agent is defined by a behavior vector, which contains the variables describing the behavior of the vehicle according to the pre-cited models (target speed, maximum acceleration, etc.).

The assumption is made that when the internal score of an agent is high (i.e. it is not bothered by other cars, see 4.2 for more details), it means that its behavior is adapted to local traffic conditions. Therefore, by transmitting its internal behavior to cars in similar traffic conditions, it may help these vehicles to improve their own internal score.

This transmission of the internal behavior is similar to the transmission of the culture defined in [14], where the behavior vector of each car has the role of the meme. Thus, it naturally leads to the use of memetic algorithms (MA) [23] as a tool to model this system.

MA are a kind of evolutionary algorithm (EA) combining a genetic algorithm (GA) with a local-search. As in GA, in memetic optimization, an initial population of solutions is generated and through a succession of phases of selection (only the best solutions are kept), mutation (existing solutions are randomly modified), and crossing-over (solutions are merged), better solutions are created. The main difference between MA and GA is that in MA the members of the population can perform an individual learning procedure capable of performing local refinements. In the present situation, we want to find which vehicles behaviors lead to a better score, therefore our population is composed of the behaviors of the vehicles, and we try to improve the score by refinements of the behaviors performed by the memetic optimization.

However, applying MA to the case of lane-change optimization requires taking some considerations into account:

- 1) the problem is continuous [24]
- 2) the evaluation of the performance is made in real-time, because the only way to evaluate the impact of the modification of the behavior of a vehicle is to let it drive for a given amount of time
- 3) the environment is continuously and rapidly changing [25]
- 4) the population size cannot be controlled
- 5) the convergence is not required: an improvement of the global score is desired, but it cannot be expected to reach the optimal solution

The optimization process is summed up in the figure 5. It is executed by each vehicle in a decentralized manner. In the following, different operators are proposed. The goal is to have a basis to evaluate the feasibility and the potential benefits of the framework, but the framework is not limited to the proposed algorithms. At fixed steps, the vehicles broadcast their score and their current behavior vector to surrounding cars, so these data can be used by vehicles to improve their self-performance.

As the problem is continuous, a real-coded genetic algorithm (RCGA) is the most appropriate basis for the genetic part of the algorithm [26].

It can be noted that this kind of on-line optimization strategy used to dynamically adapt to changing traffic conditions is similar to existing works on intersections [27]. However, our approach is different as we are focused on lane-change operations and we are using a memetic algorithm for the optimization process.

A. Behavior

$B = (\alpha_\tau, \alpha_{b_l}, \alpha_{\Delta a}, \alpha_{v_i}, \alpha_{b_r}, \alpha_{b_v}) \in [0.5, 1.5]^6$ is the vehicle behavior vector. Each α_i is a coefficient applied to the associated parameter. These factors are associated to the behavior parameters previously defined in the section 3.2. The identity vector $B_{id} = (1, 1, 1, 1, 1, 1)$ indicates that the parameters are not modified.

In terms of an evolutionary algorithm, the set of all behavior vectors (i.e. the set of vehicles) is the population and a behavior vector is a chromosome (or a meme in terms of MA).

Therefore, mutation and crossover operations are defined for the behavior vector. At fixed step, the car will apply these operations to improve its performance. A score function is also defined to evaluate the fitness of a behavior.

B. Score

Several score operators are possible according to the desired traffic improvement. In this paper, the following score operator is used:

$$s(v) = 1 + k \times \min\left(0, \frac{v - v_t}{v_t}\right) \quad (2)$$

With v the current speed, v_t the target speed, and k a strictly positive value used to scale the difference between v and v_t . When $v \rightarrow v_t$, then $s(v) \rightarrow 1$ and when $|v - v_t|$

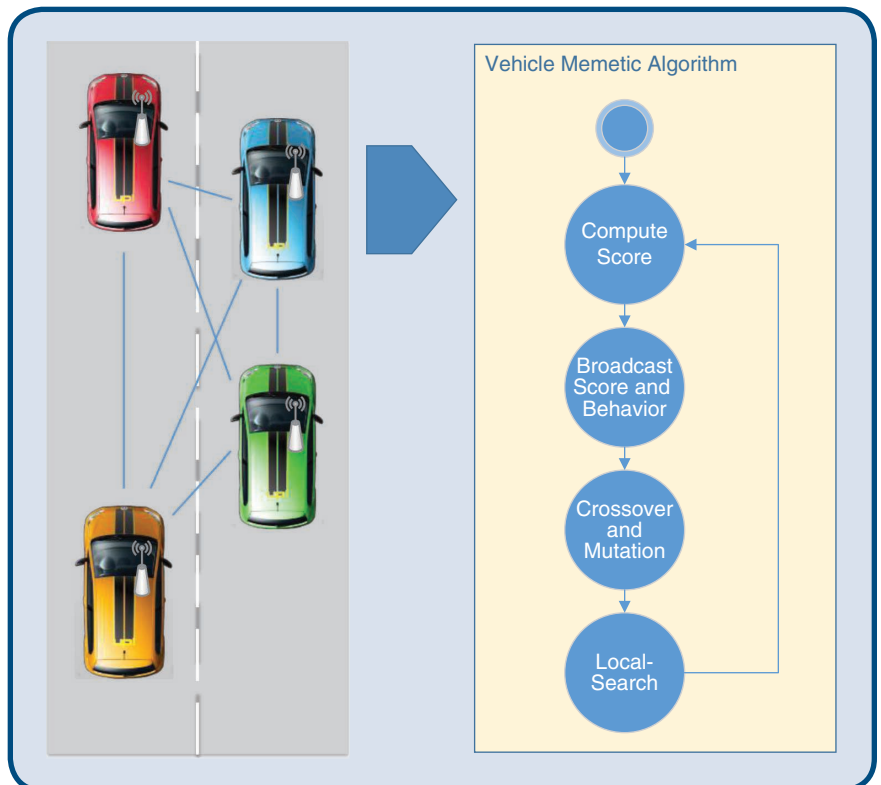


FIG 5 Memetic algorithm framework, executed by each vehicle.

increases, then $(ds/dv)(v) < 0$. The interval of definition of this operator is $[-\infty, 1]$ where 1 is the optimal score, i.e. the car is at its desired speed.

Thus, the global objective function is:

$$\text{maximize } Z = \frac{1}{N} \sum_{i=1}^N s_i \quad (3)$$

It can be noted that this function is not necessarily the criteria of the traffic that we want to improve, it is mainly a tool used to lead the evolution of the behaviors.

C. Mutation Operation

The purpose of the mutation operation is to avoid local optima by introducing randomness in the population.

In this paper the mutation operation is defined by the algorithm 2 where $\alpha_m \in [0, 1]$ is the mutation rate. This is the operator known as the **real-number creep** operator [28, 29]. A random value (uniform distribution) is generated in the interval of definition of the behavior vector component, and then a linear interpolation is made between the previous value and the generated value with α_m as the interpolation value.

D. Crossover Operation

Several crossover operators exist for RCGA [29].

If a car in a similar situation, i.e. close in distance, and has a better score, it may mean that its parameters are probably more adapted to current traffic conditions.

Therefore, to privilege behaviors with a higher fitness, the flat crossover [30] is used as a basis, but with important changes:

- 1) the influence of a parent is weighted according to its fitness

Algorithm 2 Mutation operator

```

m(B, αm) = (
    m(ατ, αm), m(αb, αm),
    m(αΔx, αm), m(αvi, αm),
    m(αb, αm), m(αbv, αm))
m(αi, αm) = lerp(αi, rand() + 0.5, αm)

```

Algorithm 3 Crossover operator

```

Get the set B10% behaviors with 10% best scores
for all (si, Bi) in B10% do
    hi = 1/2 × (α0 + 1/(1 + e-k(si-s)))
end for
B ← 1/N ∑i=1N Bihi + (1 - hi)B

```

- 2) the crossover is not made between a linear interpolation between two behaviors, but by a multilinear interpolation between a behavior and the best behaviors

In this paper the algorithm 3 is used as crossover operator. Each car selects the top 10% of received behaviors (according to their fitness), then compute a coefficient h_i for each one. The value of h_i depends on the difference between the fitness of the behaviors $s_i - s$: $\lim_{s_i - s \rightarrow -\infty} h_i = (\alpha_0/2)$ and $\lim_{s_i - s \rightarrow +\infty} h_i = (\alpha_0 + 1/2)$. $\alpha_0 \in [0, 1]$ is used to have a non-null coefficient h_i even when $s_i - s$ is low, and $k \in [0, +\infty]$ is used to scale the difference between the scores.

According to this algorithm, the weight of the modified behavior B is deliberately higher than the weight of the behaviors used for the crossover. It allows the behaviors to be more stable.

E. Local-Search Improvement

The application of a genetic algorithm is not the most efficient way to improve traffic performance. Indeed, the traffic conditions are rapidly evolving, and the same parameters may positively or negatively impact the score depending on the context, thus making it difficult to converge.

Therefore, the vehicles may use data emitted by other cars to improve their performance with a predefined algorithm.

For this purpose, the algorithm 4 is used.

This algorithm increases the target speed when the average score of surrounding cars is lower than the own score of the vehicle. If the average score of surrounding cars is higher than the own score of the vehicle, it will restore its target speed. This approach is inspired by eco-problem solving (EPS) [31], as each agent indirectly applies a pressure to surrounding agents to increase its own score. While it is called “local-search operator” to match with the definition of a memetic algorithm, it can be noted that it differs slightly from the classic definition as the purpose of this operator is not to improve directly the score of the vehicle, but indirectly by favoring the increase of the score of the surrounding vehicles.

Algorithm 4 Local-search operator

```

{ (B0, s0), (B1, s1), ..., (Bn, sn) }: Set of received broadcasted behaviors and associated score
s̄ ← 1/n ∑i=0n si
if s̄ < sself then
    αvi ← αvi + f
else
    αvi ← max(1.0, αvi - f/2)
end if

```

V Experimentation

A. Experimentation Protocol

To evaluate the performance of the framework, MAS based simulations are run.

In these simulations, vehicles are spawned from a given point on a highway with 4 lanes at a fixed rate ($r = 0.33$ veh/s), their initial speed is set to $v_i = 15 \pm 7.5$ m.s⁻¹ (uniform distribution, so the mean is 15 m.s⁻¹), and their target speed is set to $v_t = 15 \pm 7.5$ m.s⁻¹ (uniform distribution). The total length of the section used for measures is $l = 4$ km and this section begins 1 km after the spawning point of the vehicles. The total simulation time for the measures is $T_{\text{total}} = 3600$ s. The sensor ranges are up to 200 m for detection and up to 500 m for communication: these values are within the optimistic range of current sensor capabilities, but they could be reduced as long as the range of the radar is enough to guarantee safe lane-changes.

The idea that there will be such difference in the velocities of autonomous vehicles is debatable. Some argue that the use of autonomous vehicles will lead to more uniform speeds. However, for other people the increased safety induced by these vehicles will allow to increase speed limits and, thus make the traffic more heterogeneous (like it is the case on the German *Autobahn*). Indeed, in this situation the factors influencing directly the target speed are: the maximum speed of the vehicle and the comfortable speed decided by the user.

Three kind of simulations are performed:

- 1) Radar-only (illustrated in the figure 6): Without cooperative lane change, using only radar to define if a lane change is safe and thus possible; a lane-change is started as soon as the vehicle cannot reach its target speed because of a preceding vehicle, the lane-change will allow it to increase its speed (no new leader, or a new leader far enough), and the lane-change is safe (the lane-change will not make a car brake, i.e. have an acceleration below 0 m.s⁻², according to the longitudinal model).
- 2) CLC: With cooperative lane change; similar to the radar-only policy, except that communication can be used to ask for cooperation from cars in the target lane, according to the algorithm 1.

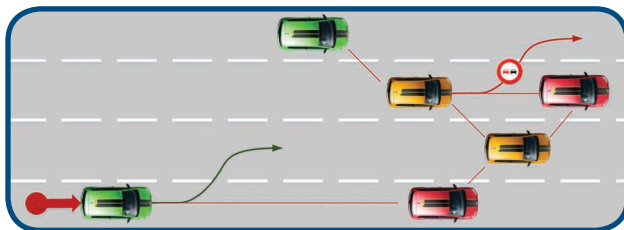


FIG 6 Schematic representation of the “radar-only” scenario: red lines shows the perceptions of each car; arrows show their intention (green if lane-change is safe, red if lane-change is not safe).

Some argue that the use of autonomous vehicles will lead to more uniform speeds. However, for other people the increased safety induced by these vehicles will allow to increase speed limits and, thus make the traffic more heterogeneous (like it is the case on the German *Autobahn*).

- 3) CLC-MA: With cooperative lane change and optimization; similar to the CLC policy, except that the behavior vector of each vehicle is regularly updated according to the framework described in the figure 5.

In the radar-only and CLC policies, the initial behavior is set to the identity behavior B_{id} and is constant. With the CLC-MA policy, the behavior is the mean of the behavior of the surrounding vehicles.

The figure 7 shows a screenshot of the dedicated simulator developed for the specific purpose of evaluating the performance of the application of the memetic optimization. In this figure, blue vehicles are not using communication, violet vehicles are emitting lane-change requests, and pink vehicles are synchronizing their speed to help lane-change operations. For the sake of readability, only a small portion of the simulated road is displayed here.

B. Parameters

For CLC-MA, the parameters defined in 3, 4 and 5 are used, respectively for the mutation algorithm, the crossover algorithm and the local search algorithm.

According to the definition of the mutation operator, a greater value of α_m means more randomness in the generation of the cars behavior. Through trial and error, the value 0.2 (see table 3) has been chosen to avoid having too much randomness.

For the crossover operator, α_0 has been set to a low value (see table 4) to limit the influence of the neighbors with low scores on the generation of new behaviors.

According to the local-search operator, a greater value for f means that the vehicles will increase their speed

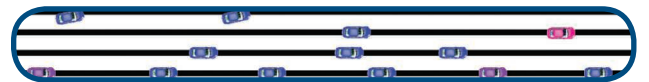


FIG 7 Screenshot of the MAS-based simulator in 2D environment used for tests.

Table 3. Mutation algorithm parameters.

Parameter	Value
α_m	0.2

quickly when the score of the neighbors is low. The value of 0.1 (see table 5) chosen here has been found by trial and error during our simulations.

The following section will present the results obtained using the presented parameters. Even though a lot of different strategies (different parameters and different operators) have been tested before getting these results, in the context of this article, the focus is not put on the calibration of the memetic algorithm but on its application to the case of lane-change. This is why there is apparently little importance given to the choice of these parameters as long as the application of the memetic optimization can give interesting results as a proof-of-concept. However, in further works several strategies could be used to find better parameters for the memetic algorithm, like those presented in [32] and [33], or we could try other strategies for the definition of operators [34].

Finally, according to the predefined framework, the parameters specific to the optimization are implied by the evolution of the traffic: the size of the population is the number of vehicles and there is no maximum generations (the optimization continue as long as there are new vehicles).

C. Results

Several runs were made, the presented figures show the average results observed. According to the figure 8, it can be seen that with radar-only and CLC policies the target

Table 4. Crossover algorithm parameters.

Parameter	Value
α_0	0.02
k	100

Table 5. Local-search algorithm parameters.

Parameter	Value
f	0.1

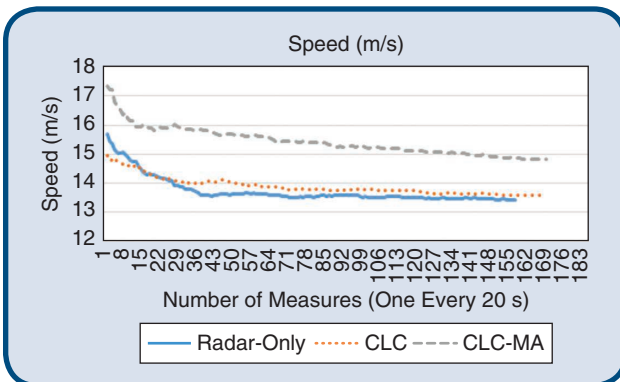


FIG 8 Mean of the average speed of the vehicles (in m.s^{-1}) for a target speed of 15 m.s^{-1} .

speed (15 m.s^{-1}) is not reached. It can be explained by the high number of vehicles whose target speeds are below 15 m.s^{-1} , and thus are slowing down the vehicles willing to go faster. At the beginning, the speed is relatively high because the fastest cars can overtake to reach their target speed, then slow cars begin to overtake slower cars, and reduce the whole traffic speed.

With CLC-MA, the average speed of the vehicles is almost equal to the average initial target speed of 15 m.s^{-1} .

When comparing the scores of the vehicles (figure 9), a similar observation can be made: the average score of the vehicles is higher using CLC-MA, compared to radar-only and CLC policies.

Finally, with CLC-MA, we can see a quick shows a considerable improvement of fluidity, according to IF (figure 10). These results were confirmed with simulations at higher speed ($v_i = v_t = 30 \pm 7.5 \text{ m.s}^{-1}$, $r = 0.33 \text{ veh/s}$) as it can be seen in the figures 11, 12 and 13: the difference is smaller but it can be explained by larger inter-vehicular distances and longer acceleration and deceleration phases, due to the increased speeds and increased range of speeds, limiting the time when the vehicles are at their target speed. However, the gain is

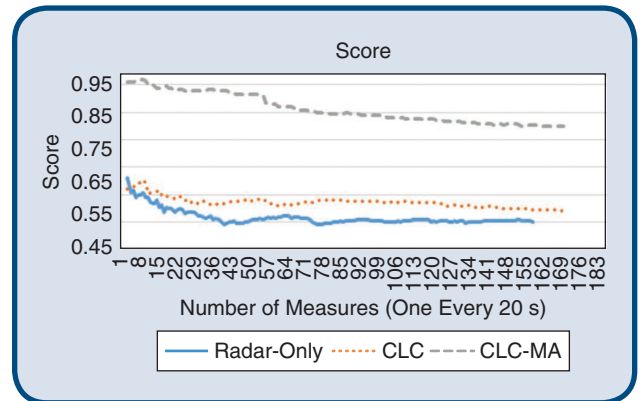


FIG 9 Average score of the vehicles for a target speed of 15 m.s^{-1} .

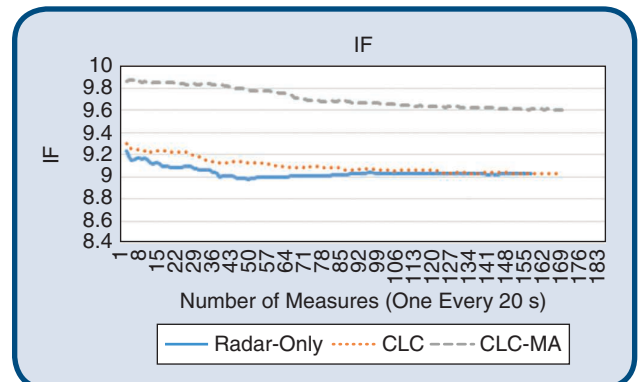


FIG 10 Average IF of the simulation in function of the chosen policy for a target speed of 15 m.s^{-1} .

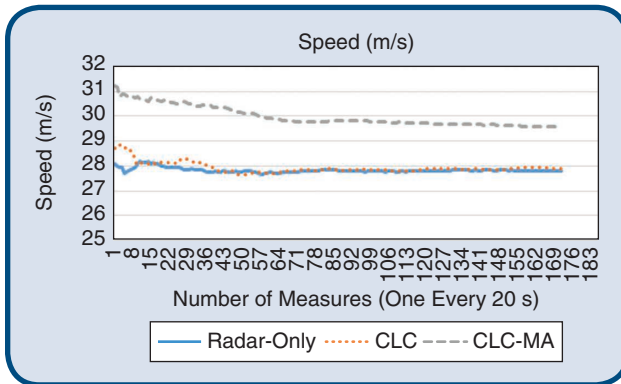


FIG 11 Mean of the average speed of the vehicles (in m.s^{-1}) for a target speed of 30 m.s^{-1} .

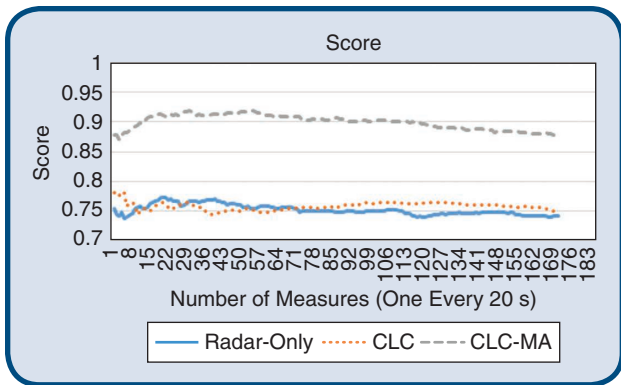


FIG 12 Average score of the simulation for a target speed of 30 m.s^{-1} in function of the chosen policy.

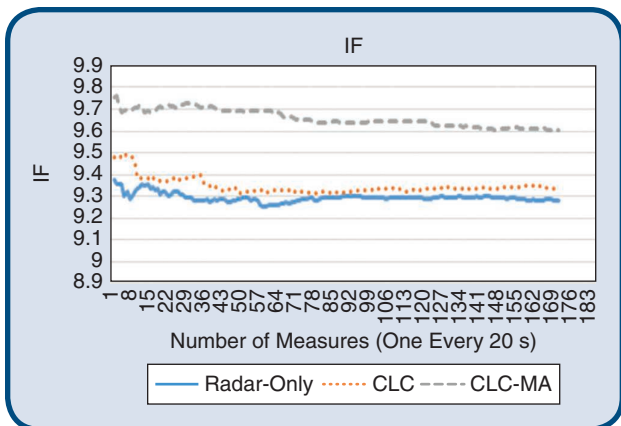


FIG 13 Average IF of the simulation for a target speed of 30 m.s^{-1} in function of the chosen policy.

Table 6. Average score for a target speed of 30 m.s^{-1} for the different policies.

Policy	Average Speed	Average Score	Average IF
Radar-only	27.83	0.752	9.294
CLC	27.93	0.758	9.347
CLC-MA	29.98	0.900	9.659

still present, especially for IF meaning that the vehicles are closer to their target speed with CLC-MA.

The average speed, score and IF for each policy in this later scenario are given by the table 6.

VI. Conclusion

This article proposes a generic framework to improve traffic performance on multi-lane roads using vehicular communication. Inspired by previous works on the cooperative intersection management, it extends them to the case of lane-change under the form of a cooperative lane-change protocol, and finally proposes a solution based on evolutionary algorithms to tune the resulting system.

Simulations are performed to estimate the potential benefits of the framework, and shows that using the presented framework can lead to noticeable gains. Thus, we introduce a proof-of-concept of the memetic optimization applied to cooperative traffic management.

In the present article, the framework is based on arbitrarily chosen models for lane-change, acceleration control, etc. and on arbitrarily defined operators for the evolutionary algorithm. These components can easily be replaced by equivalents (for instance a new lane-change model, or a new crossover operator), thus allowing to improve the overall performance of the system with few changes. Furthermore, new sets of parameters could also be chosen using the existing tuning strategies for evolutionary algorithms.

About the Authors



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